NAME:____

Math 227C Homework 6

Spring 2026
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1. Consider the chemical reaction system $A \to 2A$, $A \to \phi$, with rate parameters c_1 and c_2 respectively.
   a) Write down the master equation for this system, using the variable $\pi_i(t) = P(A(t) = i)$.
   b) Calculate explicitly the ODE for $E(X(t))$. Compare it with that for the deterministic ODE of the system
   c) Implement a few simulations for $c_1 = 10$ and either $c_2 = 9.5$ or $c_2 = 10.5$.
   d) Do you expect the master equation to have a stationary distribution? Describe the long term behavior for either case $c_1 < c_2$ and $c_1 > c_2$.

2. Suppose a coin has a probability θ of heads, where θ is unknown. An experiment involving a total of 100 coin throws leads to 75 heads and 25 tails.
   a) Implement a Bayesian algorithm to determine the posterior distribution $\pi(\theta|X = x)$ of parameter values. Assume as a prior distribution $\pi(\theta)$ the uniform distribution. b) Plot the posterior distribution (give or take a constant multiple) as a function of θ). What is a reasonable range for the value of θ)?

3. Consider the reaction $A + B \to 2B$, $B \to A$, with parameters α and β respectively. Notice the mass conservation $A + B = n$
   a) Write down the master equation of this system using the notation $\pi_j(t) = P(A(t) = j, B(t) = n - j)$.
   b) Write down the Markov chain for the stochastic system assuming $A + B = n$ for fixed n. Notice what happens when $A = n, B = 0$ in the Markov chain.
   c) What do you expect will be the long term behavior of this system given your answer in b)? Write the stationary distribution π.
   d) Implement stochastic simulations for n=10 and n=100. Compare these simulations with simulations of the deterministic ODE for this system.

4. Suppose you have expression data counts for a gene of interest. By looking at a tissue sample and modern technology, you were able to determine the number of mRNA expressed in 8 different cells during a 10 minute period of time. The protein counts are $y_1 = 120, y_2 = 150, y_3 = 100, y_4 = 90$, $y_5 = 160, y_6 = 130, y_7 = 110, y_8 = 140$. Assume that all cells have the same underlying baseline production level and that the system is modeled using a Poisson process.

a) Suppose the prior distribution for the intensity θ of the Poisson process is distributed from a gamma distribution $Gamma(a, b)$, where $a = 2, b = 0.4$. Carry out a Bayesian algorithm to determine the posterior distributions using the determined data. b) Calculate the mean and the standard deviation of θ using the posterior distribution. Note: think about what distribution θ has and use available formulas for such values. c) Consider abstract data $y_1, \ldots, y_n$ for this same problem, where these numbers are drawn from a Gaussian distribution with mean μ and variance σ^2. Calculate the expected value of θ for this data, and determine what happens to the variance as $n \to \infty$.